

Austin Bian

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EDUCATION

The Ohio State University, Columbus, OH

Bachelor of Science in Computer Science and Engineering, Minor in Mathematics

May, 2026

GPA: 3.90/4.0

WORKPLACE EXPERIENCE

Software Engineering Intern – Vectra AI, Austin, TX

May – August, 2025

- Developed Flask RESTful API for custom configuration management, used in customer deployments and internal operations within a Linux-based ecosystem.
- Designed and implemented a secure service-to-service authentication system, working with 3 cross-functional teams to architect bearer token and API key validation backed by AWS-managed secrets and role-based access controls.
- Integrated real-time API capabilities into a Django web platform through an interactive UI with live API state synchronization.
- Deployed and operated the API on EKS using Terraform modules, Helm charts, and ArgoCD GitOps workflows, backed by a PostgreSQL datastore on Aurora Serverless, with documentation covering API architecture, authentication flows, and integration patterns.

Software Engineering Intern – BAE Systems, Fairborn, OH

June – August, 2024

- Enhanced an Electron based military healthcare data integration application using TypeScript, React, HL7 protocol, and Protocol Buffers, implementing optimizations that improved data transfer rates by 50%.
- Designed and implemented a scalable testing framework using Jest and React testing library with 75% code coverage, integrated with GitLab CI/CD pipelines to automate customer quality assurance.

Integration Developer – Buckeye AutoDrive, Columbus, OH

November, 2024 – March, 2025

- Engineered a Kotlin based Android Auto app for real-time monitoring and control of an autonomous vehicle.
- Built a diagnostics module to visualize internal vehicle data sourced from processed CAN bus signals.

Tech Intern – Embrace Pet Insurance, Cleveland, OH

May – August, 2023

- Managed Active Directory, Windows OS upgrade, and inventory management projects.
- Provided technical support for Helpdesk tickets, CRM onboarding/modifications, employee onboarding processes, and network maintenance.

PROJECTS

Snapshot History Based Disk Space Analyzer

- Built a Tauri based Windows app with Rust backend + React UI for visual disk usage analysis.
- Implemented snapshot history and diffing to detect file/folder size changes across scans.
- Used SQLite for persistent storage and optimized retrieval of large snapshot datasets.
- Leveraged Rayon for multithreaded filesystem and database processing with lazy-loading and virtualization.

Thosu Application Design Project – Independent Game Development

- Developed a game from the ground up combining Shmup and precision clicking using C++ and SFML, implementing custom game engine logic like delta time loops, state management, modular rendering loops.
- Incorporated quad-tree spatial partitioning for efficient collision detection and batch rendering for particles.

LEADERSHIP & ACTIVITIES

Tau Beta Pi Honor Engineering Society – Secretary/Initiation Chair

- Planned and hosted info sessions and Initiation logistics with high engagement for more than 100 students.
- Managed the eligibility verification process, involving updating website records and reports, communication with eligible students, datasheet management, and reporting to advisors.
- Hosted an election for members interested in joining the Initiation Committee and managed approved members.

SKILLS

Languages: C++, Java, Python, TypeScript, JavaScript, C, SQL, Kotlin, HTML/CSS, Rust

Technologies & Tools: AWS, Kubernetes, Django, Flask, React, Terraform, Docker, ArgoCD, Helm, Pytorch, REST APIs, Git, GitLab CI/CD, Jest, Ansible, Linux, FastAPI, PostgreSQL